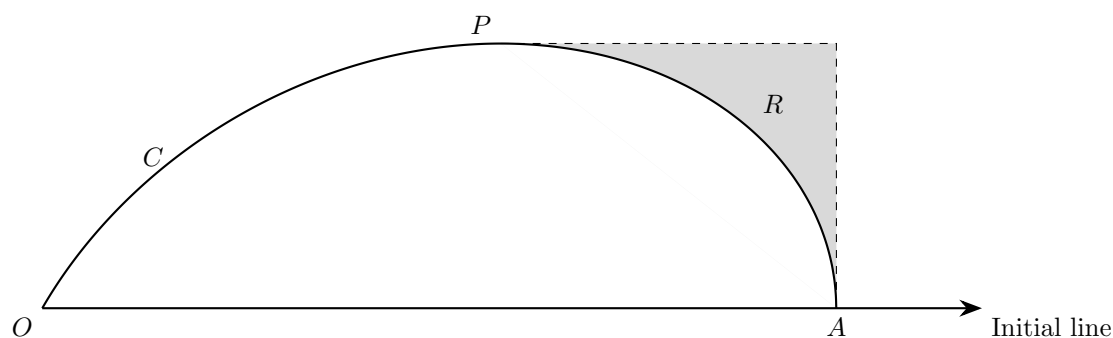


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1. The curve C shown in the figure above has polar equation

$$r = 1 + 2 \cos 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the point P on C , the tangent to C is parallel to the initial line.

- (a) Show that P has polar coordinates $\left(2, \frac{\pi}{6}\right)$. [4]

The point A on C has polar coordinates $(3, 0)$.

The finite region R , shown shaded in the figure above, is bounded by the arc AP of C , the tangent to C at P and the line through A parallel to $\theta = \frac{\pi}{2}$.

- (b) Find the exact area of R . [9]

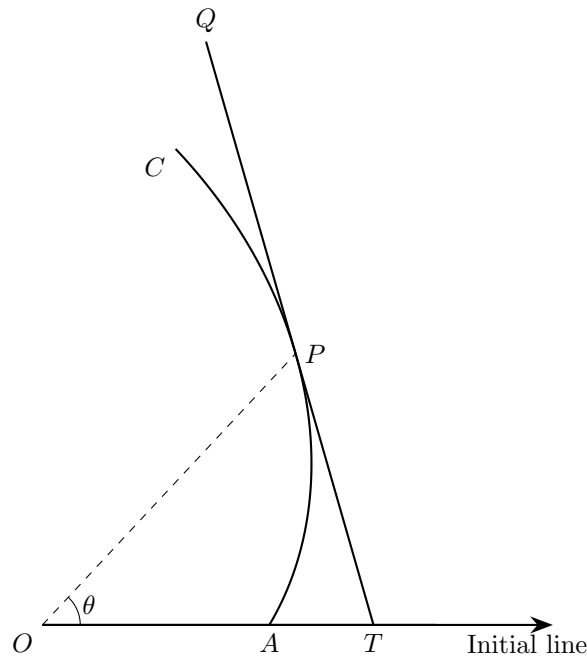
2. The curve C has polar equation

$$r = 2 \sin 3\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the non-pole point P on C , the tangent to C is parallel to the initial line.

Given that O is the pole, find the exact length of the line OP .

[7]



3. The diagram shows part of a spiral C .

The point P has polar coordinates (r, θ) , where $0 \leq \theta \leq \frac{\pi}{2}$.

The tangent at P is the line TPQ . The point A lies on C and on the initial line, with $OA = 2$.

Let ψ be the angle that TPQ makes with the initial line.

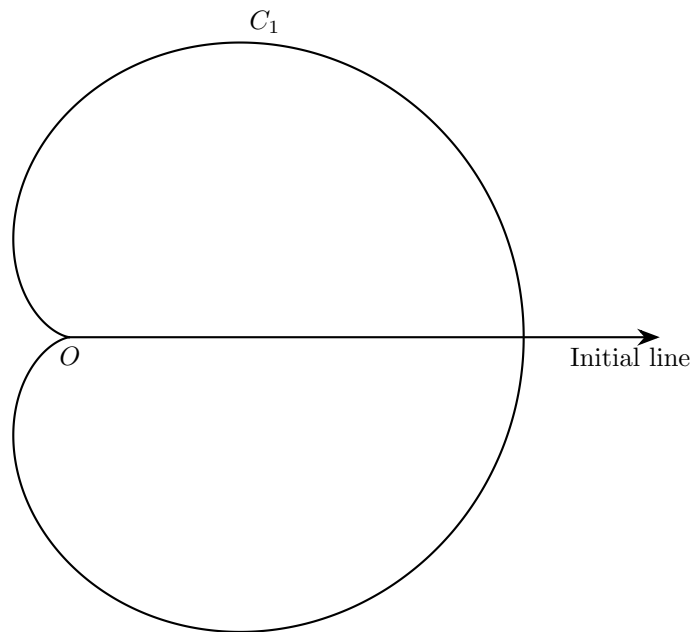
(a) By expressing the Cartesian coordinates of P in terms of r and θ , show that

$$\tan \psi = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} \quad [4]$$

(b) The spiral is equiangular: at every point P , the acute angle between OP and TPQ is a constant α , where $0 < \alpha < \frac{\pi}{2}$.

Use the result of part (a) to show that the polar equation of C is

$$r = 2e^{(\cot \alpha)\theta} \quad [7]$$



4. The diagram shows the sketch of a curve C_1 .

The polar equation of the curve C_1 is

$$r = 2(1 + \cos \theta), \quad -\pi \leq \theta \leq \pi$$

- (a) Find the area of the region bounded by the curve C_1 . [5]

- (b) The curve C_2 whose polar equation is

$$r = \sin \theta, \quad 0 \leq \theta \leq \pi$$

intersects the curve C_1 at the pole O and at the point A . The straight line drawn through A perpendicular to the initial line intersects the part of C_1 above the initial line again at the point B .

- (i) Find the polar coordinates of A . [4]
 (ii) Show that the length of OB is $\frac{6}{5}$. [6]
 (iii) Find the length of AB , giving your answer to three significant figures. [3]

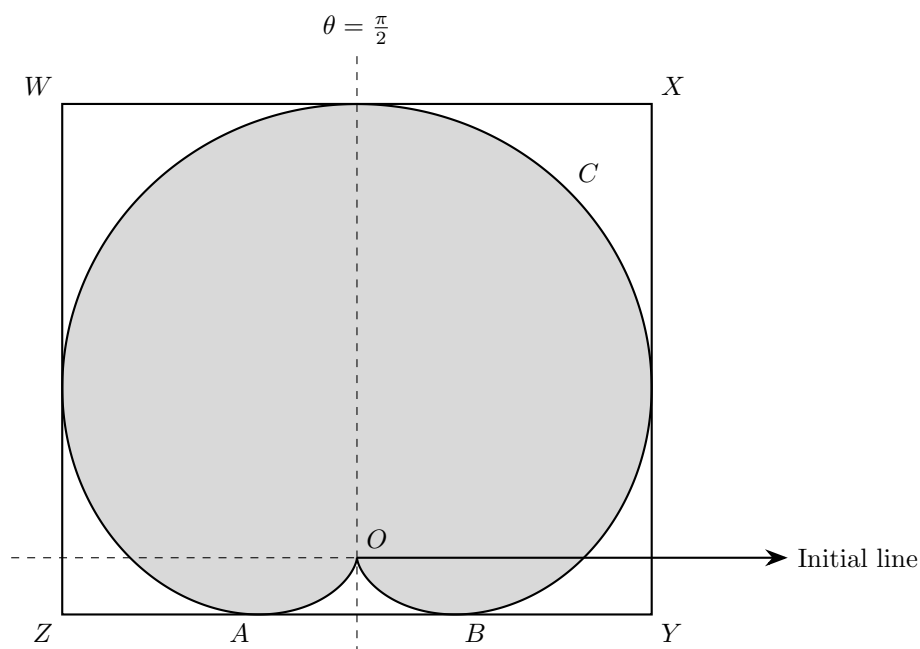
5. The curve C has polar equation

$$r = 3 \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

At the non-pole point P on C , the tangent to C is perpendicular to the initial line.

Given that O is the pole, find the exact length of the line OP .

[7]



6. The figure shows a sketch of the cardioid C with equation

$$r = a(1 + \sin \theta), \quad 0 \leq \theta < 2\pi$$

Also shown are the tangents to C that are parallel and perpendicular to the initial line. These tangents form a rectangle $WXYZ$.

(a) Find the area of the finite region, shaded in the figure, enclosed by C . [6]

(b) Find the values of θ for which the tangent to C is parallel to the initial line. Hence find the polar coordinates of the points A and B , where ZY touches C . [6]

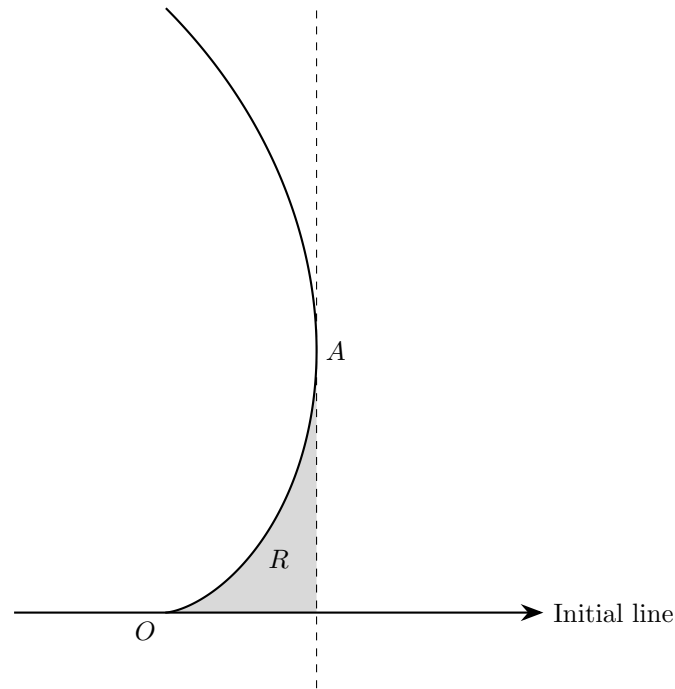
(c) Hence show that the length of WZ is $\frac{9a}{4}$. [2]

Given that the length of WX is $\frac{3\sqrt{3}a}{2}$

(d) find the area of the rectangle $WXYZ$. [2]

A logo is modelled by the cardioid C , where $a = 8$ cm. The logo is cut from the rectangular card $WXYZ$, shown in the figure.

(e) Find a numerical value for the area of card wasted in making the logo. [3]



7. The diagram above shows a sketch of the curve C with equation

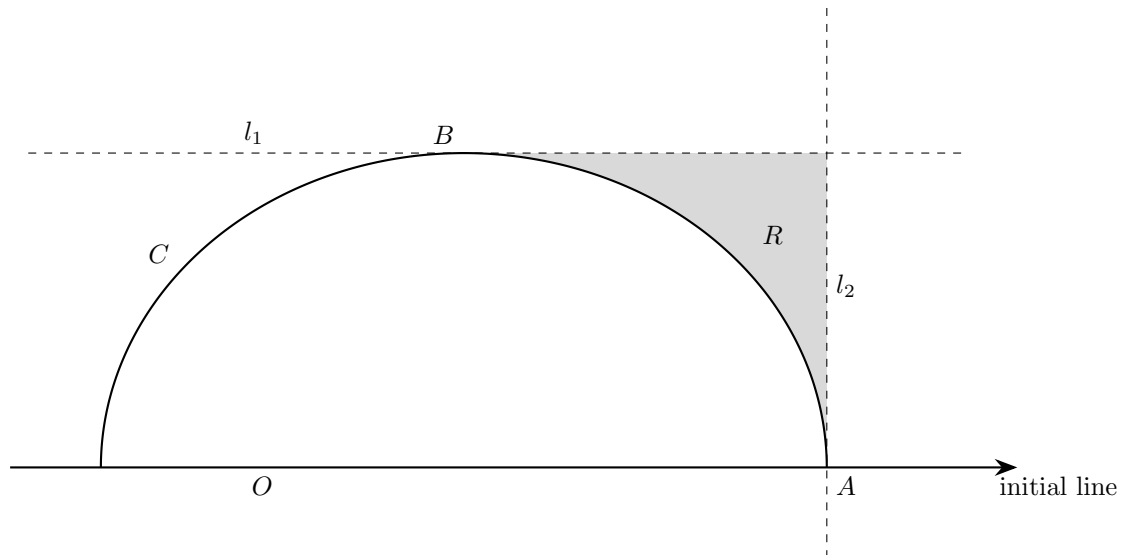
$$r = 4(1 - \cos \theta) \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The diagram above also shows the tangent to C at the point A , where A is not the pole. This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{3}\right)$. [4]

The finite region R , shown shaded in the diagram above, is bounded by C , the tangent at A and the initial line.

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(15\sqrt{3} - 8\pi)$. [6]



8. The figure above shows a sketch of the curve C with equation

$$r = \frac{6}{2 - \cos \theta} \quad 0 \leq \theta \leq \pi$$

Given that C meets the initial line at the point A , as shown in the figure above

(a) write down the polar coordinates of A . [1]

The line ℓ_1 , also shown in the figure above, is the tangent to C at the point B and is parallel to the initial line.

(b) Use calculus to determine the polar coordinates of B . [4]

The line ℓ_2 , also shown in the figure above, is the tangent to C at A and is perpendicular to the initial line.

The region R , shown shaded in the figure above, is bounded by C , ℓ_1 and ℓ_2 .

(c) By first obtaining a Cartesian equation for C , use algebraic integration to find the exact area of R , giving your answer in the form $\sqrt{3}(p + q\pi)$ where p and q are constants to be determined. [8]